

Phase 1 — Foundation

The plan: twelve lessons to a team that exists and a robot in the water. Summer 2026 · dates set with the team.

THE GOAL OF PHASE 1

Not a competition ROV. A team that exists, a starter ROV in the water, and kids who have soldered something real — a company that runs its own rituals before a single competition part is bought.

HOW PHASE 1 RUNS

- **Fixed ~3-hour lessons; the count flexes.** Phase 1 is **twelve lessons**, not six — the honest amount of time the hands-on build actually needs (MATE budgets ~24+ hours for this curriculum, six for the control box alone).
- **Numbered generically (1.1–1.12), no fixed dates.** The team is flexible on when it meets; dates are set as the phase goes.
- **MATE teaches, the parent facilitates.** MATE's narrated presentations carry the concepts as pre-watch homework; meeting time is protected for hands-on work. This is the Section 4.3-friendly pattern.
- **Every lesson ships three artifacts** — Slides, a one-page Prep Sheet, and an Instructor Cheat Sheet.

THE TWELVE LESSONS

#	LESSON	FOCUS
1.1	Kickoff	Form the company, six roles, the safety ritual, the outreach log.
1.2	Soldering Bootcamp	Safety, the three joints, the five moves, hands-on until every kid is steady.
1.3	Kit Unboxing & First Electronics	Inventory, build-sequence read, first kit joints, open the three records, capsule sketches.
1.4	Electronics Workshop	Multimeter, switches, simple circuits + lab, the Electrical Symbols handout.
1.5	Control Box I	Practice board, then start the REV 4 control-box build, meter-checking as you go.
1.6	Control Box II	Finish wiring, power wire, tether, fuse, meter-check before power-up.

#	LESSON	FOCUS
1.7	Capsule Build & First Splash	Build the capsules, pool safety, every kid in the water, capsule drop #1.
1.8	Build & Iterate I	Assemble the Triggerfish frame, thrusters, and tether.
1.9	Build & Iterate II	Pool ops, tether handling, capsule dive #2 — diagnose every leak.
1.10	Pilot & Validate I	Buoyancy, trim, every kid pilots.
1.11	Pilot & Validate II	Capsule dive #3 (gate: three dry dives), troubleshooting, first SID.
1.12	Showcase & Launch Phase 2	Demo for sponsors, retrospective, Phase 2 preview, thank-yous.

A holiday off-week falls between 1.7 and 1.8 — with flexible dates, it is simply a gap.

THE PHASE 1 EXIT GATE

Phase 1 closes — and Phase 2 begins — when all of these are true:

- The company is formed and the safety ritual runs unprompted, owned by the CEO.
- Every kid has soldered and been in the water (by 1.7).
- The SeaMATE Triggerfish is built and has been piloted by every member.
- The capsules pass three consecutive dry dives (1.11).
- Pool access is locked for the year, including the deep-water host.
- The project notebook, photo log, and outreach log are running with real entries.
- The admin track is underway with named owners.

THE PARENT LANE — SECTION 4.3, THE RULE OF THE YEAR

Teach a skill once, then step back. Parents demonstrate and photograph; they never hold the iron or route a decision. The technical documentation, marketing display, and notebook are student work, and any outside help — including AI — is disclosed. This discipline starts at the very first bench, because an adult hand on the work is the failure that costs points in April.

KEY OPEN ITEMS — TRACK AND NAG

- **Order the 12 V SLA battery, charger, and leads now** — in hand before the wiring lessons (1.5–1.6).
- **SeaMATE kits have a 2+ week lead** — order the soldering lab kits and the Simple Circuits Lab Kit early.
- **Pool prerequisites before the first splash (1.7):** lifeguard coverage, swim-ability checks, host rules in writing — confirmed before the meeting, not during it.
- **Lock the deep-water host for the year** — the Phase 2 vertical profiling float needs 2.5 m.

- **Contact Rachel McDonald, DISL** (rovcompetition@disl.edu) — confirm 2027 dates, pool depth/dimensions, required documents.
- **Admin track:** MATE registration, liability waivers, photo/media releases, medical forms, insurance proof for pool hosts, youth-protection background checks.

BUDGET — PHASE 1 LINE

Phase 1 materials run about **\$1,640**: the SeaMATE Triggerfish kit (purchased), 12 V power, two soldering stations, two multimeters and hand tools, PVC/epoxy capsule materials, the SeaMATE lab kits, and safety gear. Season build total **~\$7,841**; all-in **~\$9,800**.

THE SEASON AT THIS HONEST SCALE

PHASE	SPAN	REALISTIC ~3-HR LESSONS
1 — Foundation	summer	~12
2 — Capability	10 wks	~10–12
3 — Design	12 wks	~12–14
4 — Build	6 wks	~8–10
5 — Test & Document	4 wks	~6–8
6 — Polish & Compete	April	~4–6

SAY WATT ROBOTICS · RANGER-CLASS MATE ROV TEAM · MADISON, ALABAMA · NORTHERN GULF COAST REGIONAL, DAUPHIN ISLAND SEA LAB, APRIL 2027